# Machine Learning

## Precision, Recall, ROC and Cost-Based Decisions

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# Where this fits in our course

1. Likelihood and logistic regression: estimate model parameters.
2. Regularization: control the fitted coefficients.
3. Confusion matrix and accuracy: count the consequences of decisions.
4. Today: vary the threshold, assess ranking, and include error costs.

**Summary**

We keep the fitted model fixed throughout the worked example.

# The decision problem

A road team uses a model to decide whether to issue an impassability warning.

- $y = 1$: the road is actually impassable; $y = 0$: it is passable.
- $s$: the model score; higher means stronger evidence for class 1.
- $\widehat{y} = 1$: issue a warning; $\widehat{y} = 0$: do not warn.

Eight held-out observations are a small **teaching dataset**, not evidence for deployment.

A score can be used for ranking without being a reliable probability.

# One dataset for the whole lecture

| Observation | A | B | C | D | E | F | G | H |
|---|---|---|---|---|---|---|---|---|
| True label $y$ | 1 | 0 | 1 | 1 | 0 | 1 | 0 | 0 |
| Score $s$ | .95 | .85 | .75 | .65 | .55 | .45 | .25 | .15 |

Four actual positives: A, C, D, F.

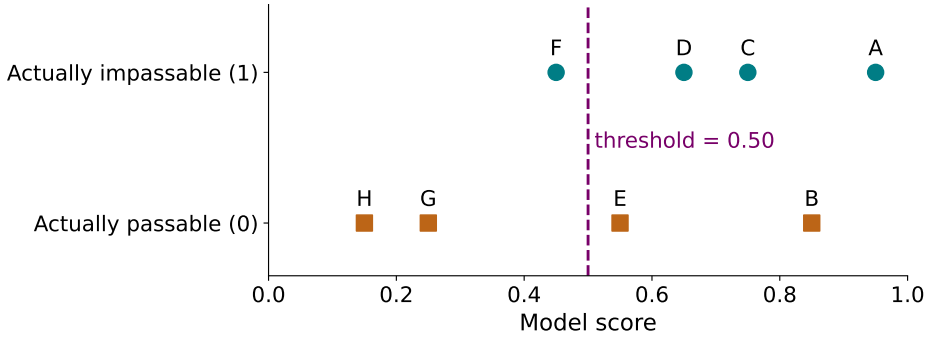
Four actual negatives: B, E, G, H.

The observations are already sorted by decreasing score.

**Notice B: a negative observation can have a high score.**

# A threshold turns scores into actions

$$\widehat{y}_i = \begin{cases} 1, & s_i \geq \tau, \\ 0, & s_i < \tau. \end{cases}$$



# Threshold 0.50: count before using formulas

| Observation | A | B | C | D | E | F | G | H |
|---|---|---|---|---|---|---|---|---|
| True label $y$ | 1 | 0 | 1 | 1 | 0 | 1 | 0 | 0 |
| Score $s$ | .95 | .85 | .75 | .65 | .55 | .45 | .25 | .15 |

Predicted positive: A, B, C, D, E.

| | Actual 1 | Actual 0 |
|---|---|---|
| Predict 1 | TP $= 3$ | FP $= 2$ |
| Predict 0 | FN $= 1$ | TN $= 2$ |

$$\text{Accuracy} = \frac{TP + TN}{8} = \frac{5}{8}.$$

# Threshold 0.40: what changed?

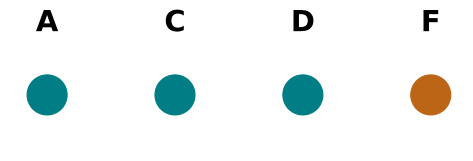
F now receives a warning. No other observation changes its predicted class.

| | Actual 1 | Actual 0 |
|---|---|---|
| Predict 1 | TP $= 4$ | FP $= 2$ |
| Predict 0 | FN $= 0$ | TN $= 2$ |

**Insight**

Lowering the threshold cannot decrease TP or FP on a fixed dataset.

# Recall step 1: start with actual positives




Keep $\tau = .50$.

Look only at the four actually impassable roads: **A, C, D, F**.

A, C and D received warnings. F was missed.

**Question: of these four actual positives, how many did we catch?**

# Recall step 2: turn that count into a rate

At $\tau = .50$, the actual-positive group contains:

$$\underbrace{A, C, D}_{TP=3} \qquad \underbrace{F}_{FN=1}.$$

Every actual positive is either caught or missed:

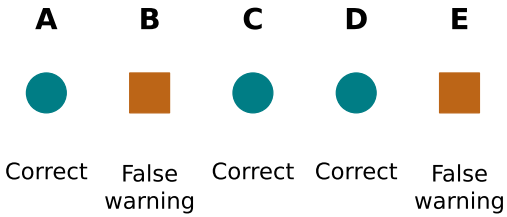
$$\text{Actual-positive count} = TP + FN = 3 + 1 = 4.$$

$$\boxed{\text{Recall} = \frac{\text{positives caught}}{\text{all actual positives}} = \frac{TP}{TP + FN} = \frac{3}{4} = 75\%.}$$

**Insight**

Recall asks how much of the real problem we detected.

# Precision step 1: start with issued warnings




Still use $\tau = .50$.

Now look only at the five warned roads: **A, B, C, D, E**.

A, C and D are actually impassable. B and E are false warnings.

**Question: of these five warnings, how many were correct?**

# Precision step 2: turn that count into a rate

At $\tau = .50$, the warned group contains:

$$\underbrace{A, C, D}_{TP=3} \qquad \underbrace{B, E}_{FP=2}.$$

Every warning is either correct or false:

$$\text{Number of warnings} = TP + FP = 3 + 2 = 5.$$

$$\boxed{\text{Precision} = \frac{\text{correct warnings}}{\text{all warnings}} = \frac{TP}{TP + FP} = \frac{3}{5} = 60\%.}$$

**Insight**

Precision asks how trustworthy the issued warnings were.

# The numerator is the same; the denominator changes

| | Recall | Precision |
|---|---|---|
| Start with | Actual positives | Predicted positives |
| In our example | A, C, D, F | A, B, C, D, E |
| Count successes | A, C, D | A, C, D |
| Calculate | $3/4 = 75\%$ | $3/5 = 60\%$ |

**Recall:** did we find the impassable roads?

**Precision:** were our warnings usually correct?

A missed event affects recall. A false warning affects precision.

# Threshold step 1: a false warning lowers precision

At $\tau = .95$, only A receives a warning.

At $\tau = .85$, B also receives a warning. B is actually negative.

| Threshold | Warned roads | Recall | Precision |
|---|---|---|---|
| .95 | A | $1/4$ | $1/1 = 1$ |
| .85 | A, B | $1/4$ | $1/2$ |

**Insight**

We added no detected positive, but did add a false warning. Recall stays fixed; precision falls.

# Threshold step 2: correct warnings raise precision

Lower the threshold from .85 to .65.

C and D now receive warnings; both are actual positives.

| Threshold | Warned roads | Recall | Precision |
|---|---|---|---|
| .85 | A, B | 1/4 | 1/2 |
| .65 | A, B, C, D | 3/4 | 3/4 |

**Insight**

Both recall and precision increased. Lowering a threshold does not force precision to decrease at every step.

# Threshold step 3: recover the last missed positive

At .50, the model warns about A, B, C, D, E.

At .40, it also warns about F, an actual positive.

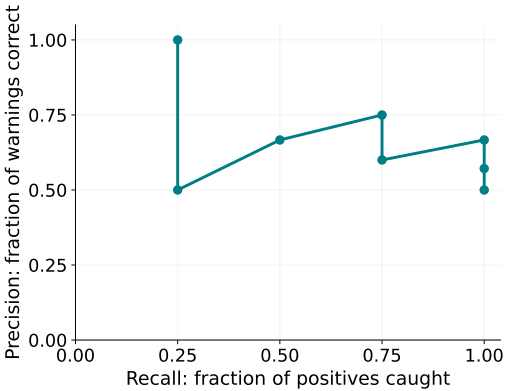
| Threshold | TP | FP | Recall | Precision |
|---|---|---|---|---|
| .50 | 3 | 2 | $3/4$ | $3/5$ |
| .40 | 4 | 2 | $4/4 = 1$ | $4/6 = 2/3$ |

**Recall is now 100%, but not every warning is correct.**

B and E are still false warnings.

If we warn about all eight roads, recall stays 1 and precision falls to $4/8$.

# Build a precision-recall plot




Each threshold gives a point:

$$(\text{Recall}, \text{Precision}).$$

The same eight observations produce this plot.

Adding a negative: recall stays fixed, precision falls.
Adding a positive: recall rises and precision can rise.

With no warnings, precision is $0/0$: undefined. That endpoint is omitted here.

# Does a PR curve tell us how good a model is?

Compare two models on the same population: both find 80 of 100 actual positives.

| At recall 0.80, with 100 actual positives | Model A | Model B |
|---|---|---|
| Precision | .80 | .50 |
| Total warnings: $TP/\text{Precision}$ | 100 | 160 |
| False warnings | 20 | 80 |

**Insight**

At the same recall, higher precision means fewer false warnings for the same number of positives found.

# Why precision changes with prevalence

Let $\pi$ be the fraction of observations that are actually positive.

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{\pi\,\text{TPR}}{\pi\,\text{TPR} + (1-\pi)\text{FPR}}.$$

For the same TPR $= .90$ and FPR $= .10$:

| Prevalence $\pi$ | Calculation | Precision |
|---|---|---|
| .50 | $.45/(.45 + .05)$ | .90 |
| .10 | $.09/(.09 + .09)$ | .50 |

Fewer actual positives can mean a larger fraction of warnings are false, even when TPR and FPR are unchanged.

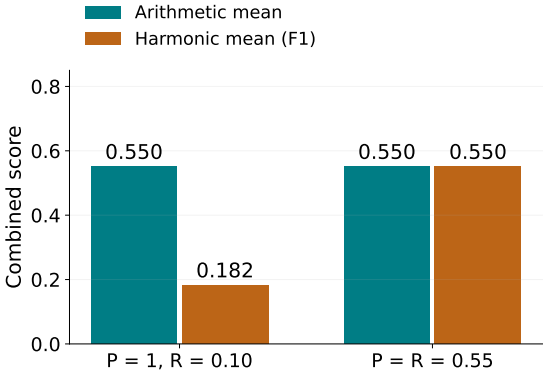
# Reading a PR curve responsibly

- Higher precision at a given recall is preferable; higher recall at a given precision is preferable.
- If curves cross, the better model depends on the operating region and consequences.
- For selection independent of the label, population precision equals prevalence $\pi$, not always .50.
- A high PR curve does not establish calibrated probabilities or choose a cost-optimal threshold.

Warning about everyone gives recall 1 and precision $\pi$.

Finite-sample random rankings fluctuate around the population reference.

# Why not use the arithmetic mean?




Policy A: precision 1, recall .10.
All warnings are correct, but 90% of actual positives are missed.

$$\mathrm{AM} = \frac{1 + .10}{2} = .55.$$

Policy B: precision $=$ recall $=$ .55.
Its arithmetic mean is also .55.

**AM treats the two pairs equally. F1 penalizes the imbalance.**

# Why the harmonic mean penalizes imbalance

For positive precision $P$ and recall $R$:

$$\mathrm{HM} = \frac{2}{1/P + 1/R} = \frac{2PR}{P + R}.$$

A small value has a large reciprocal, pulling the mean downward.

$$P = 1,\ R = .10: \quad \mathrm{HM} = \frac{.20}{1.10} \approx .182.$$

For the balanced pair $P = R = .55$, HM = AM = .55.

$$\mathrm{AM} - \mathrm{HM} = \frac{(P - R)^2}{2(P + R)} \geq 0.$$

**HM is a design choice, not a requirement.** F1 treats P and R symmetrically; use explicit costs when error consequences are known.

# F1 combines precision and recall

The harmonic mean is

$$F_1 = \frac{2\,\mathrm{Precision}\,\mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}}.$$

Substitute $TP/(TP + FP)$ and $TP/(TP + FN)$, then cancel:

$$F_1 = \frac{2TP}{2TP + FP + FN}.$$

At $\tau = .50$:

$$F_1 = \frac{2(3)}{2(3) + 2 + 1} = \frac{6}{9} = \frac{2}{3}.$$

F1 excludes true negatives and does not encode arbitrary error costs.

# Rates use different reference groups

$$\mathrm{TPR} = \mathrm{Recall} = \frac{TP}{TP + FN}$$

Of all actually impassable roads, what fraction received a warning?

$$\mathrm{FPR} = \frac{FP}{FP + TN}$$

Of all actually passable roads, what fraction received a warning? **Specificity**

$= TN/(FP + TN) = 1 - \mathrm{FPR}$.

These denominators count actual classes, not predicted classes.

# Each threshold gives one ROC point

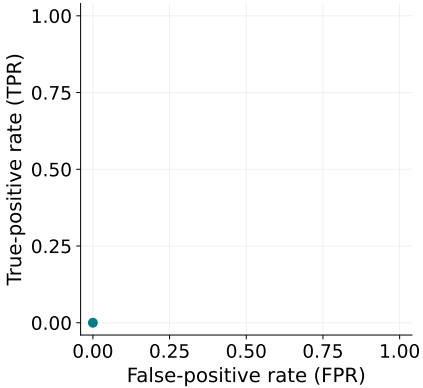
| Threshold | TP | FP | FPR | TPR |
|---|---|---|---|---|
| .50 | 3 | 2 | .50 | .75 |
| .40 | 4 | 2 | .50 | 1.00 |

The horizontal coordinate is **FPR**; the vertical coordinate is **TPR**.

An ROC curve collects these coordinates as the threshold varies.

ROC stands for **receiver operating characteristic**.

# ROC step 0: warn nobody



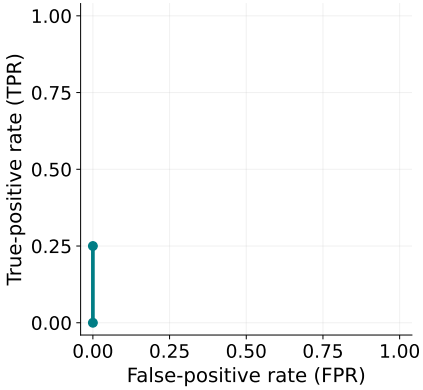

Choose $\tau > 0.95$.

$$TP = 0, \quad FP = 0.$$

$$(\mathrm{FPR}, \mathrm{TPR}) = (0, 0).$$

All four actual positives are missed.

# ROC step 1: include A



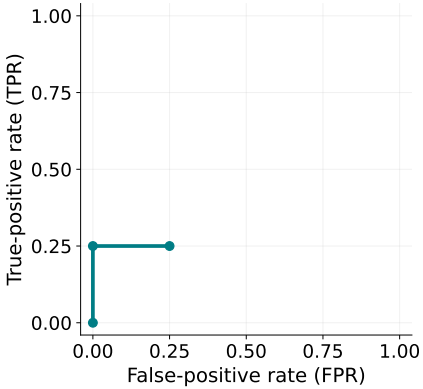

Lower $\tau$ to 0.95.
A has true label 1.

$$TP : 0 \to 1, \quad FP = 0.$$

$$(\mathrm{FPR}, \mathrm{TPR}) = (0, \tfrac{1}{4}).$$

Crossing a positive moves **up**.

# ROC step 2: include B



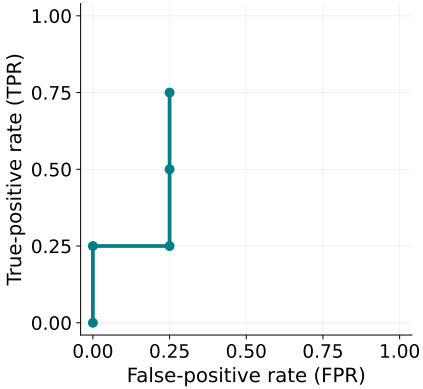

Lower $\tau$ to 0.85.
B has true label 0.

$$TP = 1, \quad FP : 0 \to 1.$$

$$(\mathrm{FPR}, \mathrm{TPR}) = (\tfrac{1}{4}, \tfrac{1}{4}).$$

Crossing a negative moves **right**.

# ROC steps 3 and 4: include C and D

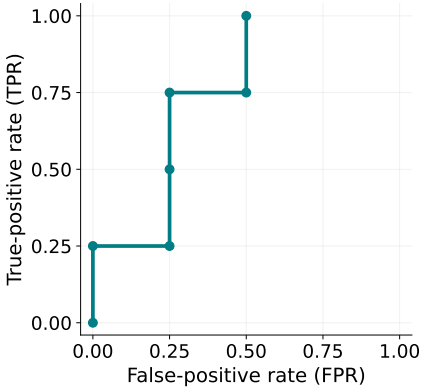



At $\tau = .75$: $(\frac{1}{4}, \frac{2}{4})$.

At $\tau = .65$: $(\frac{1}{4}, \frac{3}{4})$.

The false-positive count stays at 1.
Two more positives have received warnings.

# ROC steps 5 and 6: include E and F

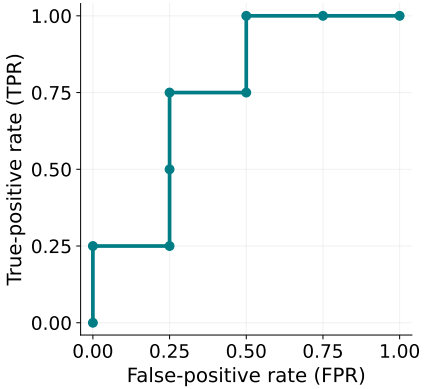



E is negative: move right.
At $\tau = .55$: $(\frac{2}{4}, \frac{3}{4})$.

F is positive: move up.
At $\tau = .45$: $(\frac{2}{4}, 1)$.

All positives are now detected, but two false warnings remain.

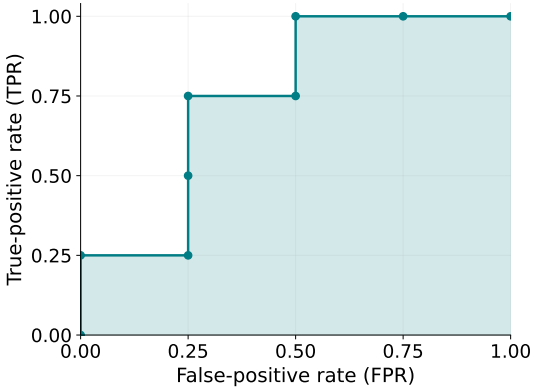
# The complete ROC curve




| $\tau$ | TP | FP | FPR | TPR |
|---|---|---|---|---|
| 1.01 | 0 | 0 | 0.00 | 0.00 |
| 0.95 | 1 | 0 | 0.00 | 0.25 |
| 0.85 | 1 | 1 | 0.25 | 0.25 |
| 0.75 | 2 | 1 | 0.25 | 0.50 |
| 0.65 | 3 | 1 | 0.25 | 0.75 |
| 0.55 | 3 | 2 | 0.50 | 0.75 |
| 0.45 | 4 | 2 | 0.50 | 1.00 |
| 0.25 | 4 | 3 | 0.75 | 1.00 |
| 0.15 | 4 | 4 | 1.00 | 1.00 |

G and H move right to $(1, 1)$.

# AUC: area under the ROC curve



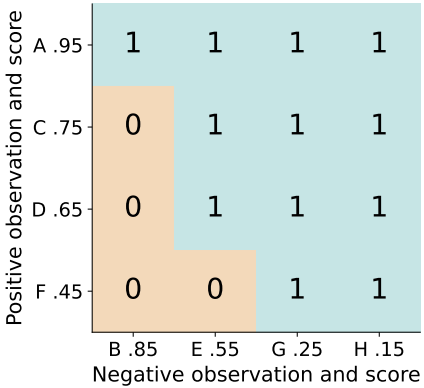

Four strips each have width $1/4$. Their heights are $1/4, 3/4, 1, 1$.

$$\mathrm{AUC} = \tfrac{1}{4}\left(\tfrac{1}{4} + \tfrac{3}{4} + 1 + 1\right)$$

$$= \frac{12}{16} = 0.75.$$

Vertical segments contribute no area.

# AUC: compare a positive with a negative




There are $4 \times 4 = 16$ pairs.

Entry 1: the positive has the higher score.
Entry 0: the negative has the higher score.

12 correctly ordered pairs:

$$\mathrm{AUC} = 12/16.$$

# The general AUC formula

Let $s_+$ be the score of a randomly selected positive and $s_-$ that of a randomly selected negative.

$$\mathrm{AUC} = P(s_+ > s_-) + \tfrac{1}{2}P(s_+ = s_-).$$

In a finite sample:

$$\mathrm{AUC} = \frac{\text{wins} + \frac{1}{2}\text{ties}}{n_+ n_-}.$$

$n_+$ and $n_-$ are the positive and negative counts.

AUC measures ranking. It is **not** the fraction of correct classifications.

# What AUC does not tell us

- Which threshold should we use? AUC does not select one.
- How costly are the errors? AUC does not specify their costs.
- Is a score of .80 really an 80% probability? AUC does not check this.

If every score becomes $s^2$, the ranking and AUC stay the same.

But using the same numerical threshold can change the decisions.

**Summary**

Ranking, probability calibration, and decision quality are different.

# State the costs before choosing a threshold

For this teaching example, costs are measured in agreed decision units:

| Action | Actually impassable | Actually passable |
|---|---|---|
| Issue warning | 0 | $C_{FP} = 2$ |
| Do not warn | $C_{FN} = 6$ | 0 |

A missed impassable road costs three times a false warning.

These illustrative costs are assumptions, not measured safety valuations.

# Compare the costs of two policies

$$\text{Total cost} = 2FP + 6FN.$$

| Threshold | FP | FN | Calculation | Cost |
|---|---|---|---|---|
| .60 | 1 | 1 | $2(1) + 6(1)$ | 8 |
| .40 | 2 | 0 | $2(2) + 6(0)$ | 4 |

Both have accuracy 6/8, but their costs differ.

**Choose .40 among these two candidates under the stated costs.**

This comparison does not prove .40 is universally optimal.

# A probability lets us compare actions directly

For a new road with features $x$, suppose

$$p = P(Y = 1 \mid x)$$

is a reliable probability for the deployment population.

We do not yet know the road's true state.

So compare the **expected cost** of each action.

Here, correct decisions cost zero, a false warning costs 2, and a missed event costs 6.

# Expected cost of issuing a warning

If we predict 1:

- With probability $p$, the road is impassable: cost 0.
- With probability $1 - p$, the road is passable: cost $C_{FP}$.

$$R(1 \mid x) = 0 \cdot p + C_{FP}(1 - p).$$

$$\boxed{R(1 \mid x) = C_{FP}(1 - p)}$$

$R(1 \mid x)$ means expected cost of action 1 for this road.

# Expected cost of not issuing a warning

If we predict 0:

- With probability $p$, the road is impassable: cost $C_{FN}$.
- With probability $1 - p$, the road is passable: cost 0.

$$R(0 \mid x) = C_{FN}p + 0 \cdot (1 - p).$$

$$\boxed{R(0 \mid x) = C_{FN}p}$$

Choose the action with the smaller expected cost.

# Derive the probability threshold

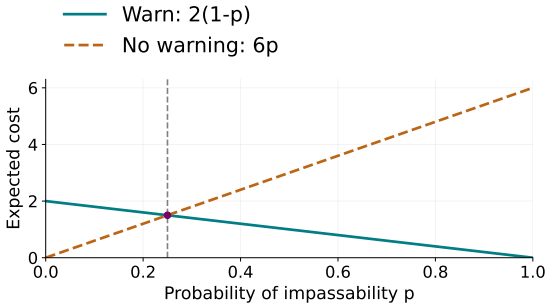
Choose action 1 when $R(1 \mid x) \leq R(0 \mid x)$:

$$\begin{aligned} C_{FP}(1-p) &\leq C_{FN}p, \\ C_{FP} &\leq (C_{FP}+C_{FN})p, \\ p &\geq \frac{C_{FP}}{C_{FP}+C_{FN}}. \end{aligned}$$

$$\boxed{\tau^* = \frac{2}{2+6} = \frac{1}{4}}$$

At equality, either action has the same expected cost; our rule chooses 1.

# The threshold is where the two costs cross




Below $p = .25$: not warning is cheaper.

Above $p = .25$: warning is cheaper.

Equal error costs would give .50.
A larger missed-event cost lowers the threshold.

# Why not just pick the best sample threshold?

The analytic rule .25 minimizes expected cost under the probability assumptions.

On our eight observations, threshold .25 gives FP $= 3$, FN $= 0$: cost 6. Threshold .40 gives cost 4.

There is no contradiction:

- Eight realized outcomes are a noisy sample, not expected costs.
- The toy scores have not been shown to be calibrated probabilities.

Use validation data to choose a practical policy; reserve test data for evaluation.

# Population composition changes total cost

Let $\pi = P(Y = 1)$ be the fraction of actually impassable roads in a population of size $N$.

Expected actual negatives: $N(1 - \pi)$.

Expected false positives: $N(1 - \pi)\mathrm{FPR}$.

Expected actual positives: $N\pi$.

Expected false negatives: $N\pi(1 - \mathrm{TPR})$.

We assume the same class-conditional TPR and FPR apply in this population.

# Derive cost per observation

Multiply each expected error count by its cost:

$$\text{Expected total cost} = C_{FP} N(1-\pi)\mathrm{FPR} + C_{FN} N\pi(1-\mathrm{TPR}).$$

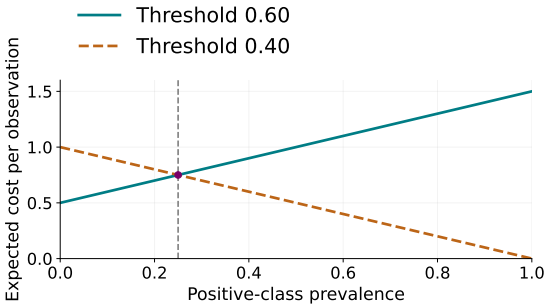
Divide by $N$:

$$\boxed{R = C_{FP}(1-\pi)\mathrm{FPR} + C_{FN}\pi(1-\mathrm{TPR})}.$$

This averages over both actual classes.

It explains why the same TPR and FPR can have different practical consequences in different populations.

# Compare policies as prevalence changes




With costs 2 and 6:

$$
\begin{aligned}
R_{.60} &= 2(1-\pi)\tfrac{1}{4} + 6\pi\tfrac{1}{4} \\
&= .5 + \pi, \\
R_{.40} &= 2(1-\pi)\tfrac{1}{2} + 0 \\
&= 1 - \pi.
\end{aligned}
$$

Tie when $.5 + \pi = 1 - \pi$: $\pi = .25$.
Below .25 choose .60; above .25 choose .40, among these two policies.

# Extension: equal-cost lines on the ROC plot

Write $u = \mathrm{FPR}$ and $v = \mathrm{TPR}$. For a fixed expected cost $r$:

$$r = C_{FP}(1-\pi)u + C_{FN}\pi(1-v).$$

Rearrange, for positive costs and $0 < \pi < 1$:

$$v = 1 - \frac{r}{C_{FN}\pi} + \frac{C_{FP}(1-\pi)}{C_{FN}\pi}u.$$

Equal-cost lines have slope $C_{FP}(1-\pi)/(C_{FN}\pi)$.

For a fixed FPR, a higher TPR means lower cost.

For an empirical ROC, compare achievable points; a smooth tangency is not required.

# Training weights and decision thresholds differ

**Weighted training** changes the objective used to fit parameters:

$$-\sum_i \left[ w_1 y_i \log p_i + w_0 (1 - y_i) \log(1 - p_i) \right].$$

$w_1, w_0 > 0$ weight positive and negative training examples.

**Threshold selection** changes the action rule for an already fitted model.

Class weights do not automatically turn its outputs into deployment probabilities.

Do not assume weighting and lowering the threshold are interchangeable in a fitted, regularized model.

# A practical evaluation protocol

1. Fit preprocessing and the model using training data.
2. Use validation data to compare models and choose a threshold; assess calibration if probabilities drive costs.
3. Freeze the model, preprocessing and decision rule.
4. Report test-set confusion counts, TPR, FPR, AUC and the stated cost.

Report sample size, class prevalence and the positive-class definition.

An eight-observation teaching example cannot establish deployment reliability.

# Check the reasoning, not just the formula

1. Does lowering a threshold retrain the model?
2. Why does a negative observation move ROC rightward?
3. Can two thresholds have equal accuracy but different costs?
4. Does a high AUC guarantee good probability estimates?
5. If missed positives become more costly, which way should the probability threshold move?

# The formulas and what they answer

$$\mathrm{TPR} = \frac{TP}{TP + FN}, \qquad \mathrm{FPR} = \frac{FP}{FP + TN}$$

$$\mathrm{AUC} = \frac{\text{wins} + \frac{1}{2}\text{ties}}{n_+ n_-}$$

$$\text{Observed cost} = C_{FP} FP + C_{FN} FN.$$

$$\tau^* = \frac{C_{FP}}{C_{FP} + C_{FN}} \quad \text{for calibrated probabilities and zero correct-decision costs.}$$

$$R = C_{FP}(1 - \pi)\mathrm{FPR} + C_{FN}\pi(1 - \mathrm{TPR}).$$

**Count errors; assess ranking; choose actions using consequences.**